

L'ETAPE, QC, Canada

WMO: 713820

Lat: 47.5622N

Lon: 71.2289W

Elev: 791

StdP: 92.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.9	-30.1	-36.9	0.1	-32.8	-33.7	0.2	-30.0	7.6	-7.9	7.0	-10.0	1.3	10	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	24.6	17.0	22.9	16.0	21.3	15.4	19.0	21.9	18.0	20.7	17.0	19.7	3.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	14.2	20.2	17.0	13.3	19.3	16.0	12.5	18.3	57.5	21.9	53.9	20.8	50.8	19.7	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.8	5.2	-38.0	27.4	3.4	1.8	-40.4	28.7	-42.4	29.7	-44.3	30.7	-46.7	32.0
			-38.2	20.7	3.4	1.2	-40.6	21.5	-42.6	22.2	-44.5	22.9	-47.0	23.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-16.1	-14.8	-9.4	-1.7	6.0	11.8	14.5	13.4	9.6	2.9	-4.2	-11.3
	DBStd	11.99	7.63	7.09	6.96	5.20	4.10	3.87	2.94	3.07	4.09	4.60	5.50	6.72
	HDD10.0	3992	809	693	603	352	135	24	2	6	57	223	427	660
	HDD18.3	6651	1068	927	861	601	381	198	122	155	263	478	677	918
	CDD10.0	392	0	0	0	0	12	78	143	112	44	4	0	0
	CDD18.3	10	0	0	0	0	0	2	5	2	1	0	0	0
	CDH23.3	114	0	0	0	0	4	42	42	18	7	0	0	0
	CDH26.7	8	0	0	0	0	0	4	3	0	1	0	0	0
Wind	WSAvg	2.3	2.3	2.3	2.5	2.5	2.3	2.2	2.2	2.1	2.3	2.4	2.5	2.3
Precipitation	PrecAvg	1300	77	73	96	86	105	116	143	113	133	123	115	111
	PrecMax	1551	132	171	152	165	171	201	211	260	275	230	172	200
	PrecMin	1104	23	35	46	42	40	66	44	47	52	46	33	39
	PrecStd	121	32	34	33	33	38	35	42	49	47	40	34	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	3.7	9.2	15.0	23.5	27.1	26.8	25.3	24.2	18.2	11.5	5.0
		MCWB	5.4	2.4	4.1	8.3	13.5	17.8	18.2	17.7	17.4	12.6	9.8	4.7
	2%	DB	1.2	1.0	5.1	11.3	20.3	24.3	24.3	23.4	21.0	15.0	7.9	1.7
		MCWB	0.9	0.2	2.1	6.2	11.4	16.1	17.3	16.7	16.1	12.2	6.7	1.3
	5%	DB	-2.0	-1.4	2.9	8.6	17.4	22.1	22.6	21.6	18.9	12.5	5.4	0.1
		MCWB	-2.4	-2.3	0.6	4.3	10.7	14.9	16.3	16.0	15.3	9.8	4.3	-0.2
	10%	DB	-4.8	-3.5	0.9	6.1	14.7	19.8	21.0	19.9	16.9	10.2	3.0	-1.9
		MCWB	-5.3	-4.3	-0.5	2.7	9.6	14.0	15.9	15.3	14.1	8.5	2.2	-2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	3.1	6.5	10.3	16.4	19.7	20.6	19.6	19.2	14.4	10.7	4.7
		MCDB	5.5	3.4	7.5	12.9	20.1	24.4	23.9	22.7	21.9	15.6	11.0	4.9
	2%	WB	0.9	0.3	3.0	7.6	14.0	18.0	19.2	18.3	17.6	12.4	6.9	1.3
		MCDB	1.0	0.9	4.4	10.1	17.0	21.9	21.7	20.9	19.5	14.6	7.4	1.4
	5%	WB	-2.4	-2.3	1.0	4.8	11.8	16.6	18.0	17.3	16.1	10.4	4.5	-0.4
		MCDB	-2.1	-1.6	2.5	7.4	14.8	19.9	20.7	19.9	18.1	11.9	5.2	-0.1
	10%	WB	-5.2	-4.3	-0.5	3.1	10.2	15.3	17.0	16.4	14.6	8.5	2.2	-2.5
		MCDB	-4.9	-3.5	1.1	5.6	14.2	18.4	19.7	18.9	16.5	9.9	2.9	-2.1
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.4	12.5	10.6	11.8	11.9	10.6	10.7	10.0	7.8	7.3	9.1
		MCDBR	11.9	12.3	14.0	14.3	17.8	15.9	13.7	13.7	13.2	12.7	10.2	8.9
		MCWBR	11.8	11.1	11.1	9.4	10.1	8.6	7.5	7.9	8.4	8.7	8.7	9.2
	5% WB	MCDBR	11.5	11.0	11.3	11.7	13.3	12.4	10.5	10.4	10.8	10.7	9.2	9.2
		MCWBR	11.5	10.3	9.5	8.7	9.3	7.8	6.7	7.9	8.4	8.7	8.7	9.4
Clear-Sky Solar Irradiance	taub	0.242	0.255	0.287	0.319	0.329	0.349	0.364	0.346	0.333	0.307	0.283	0.251	
	taud	2.333	2.289	2.238	2.334	2.484	2.432	2.390	2.447	2.484	2.545	2.490	2.363	
	Ebn at Noon	860	921	935	934	934	913	892	897	879	855	805	800	
	Edh at Noon	70	92	114	116	104	111	114	103	90	73	62	61	
All-Sky Solar Radiation	RadAvg	1.36	2.28	3.44	4.44	5.06	5.42	5.23	4.54	3.53	2.08	1.18	0.93	
	RadStd	0.17	0.28	0.27	0.36	0.53	0.37	0.55	0.30	0.28	0.18	0.13	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page